

- Laptop Hub: Project Progress Report (Phase 3)
 - Phase 9: User Engagement & Personalization
 - 1. Wishlist Functionality
 - Phase 10: Intelligent AI Integration
 - 1. Gemini AI Product Recommendations
 - Phase 11: UI/UX Enhancements & Homepage Modernization
 - 1. Dynamic Banner Slider
 - Phase 12: Profile Decoupling & Security Hardening
 - 1. Modular Profile Components
 - Phase 13: Seller Portal & Product Deletion
 - 1. Authorized Product Deletion
 - 2. Seller Auction Management
 - Current Status

Laptop Hub: Project Progress Report (Phase 3)

This report summarizes the third phase of technical milestones and feature implementations for the **Laptop Hub** platform, focusing on user engagement, AI integrations, UI/UX polish, profile decoupling, and advanced seller tools.

Phase 9: User Engagement & Personalization

1. Wishlist Functionality

- **Wishlist Service:** Implemented `wishlist-service.ts` to manage wishlist interactions (add, remove, and retrieve) using relational database queries.
 - **Dynamic UI Bindings:** Integrated wishlist buttons/icons with optimistic state updates on `ProductCard`, `AuctionCard`, and product detail views.
 - **Dedicated Profile Tab:** Developed a custom `WishList` component within the profile section where users can browse and manage their saved listings.
-

Phase 10: Intelligent AI Integration

1. Gemini AI Product Recommendations

- **AI Recommendation API:** Developed `api/recommend/route.ts` utilizing the **Gemini AI SDK** to intelligently suggest relevant laptops based on the user's intent when a search returns zero results.
 - **Interactive UI Panel:** Created the `AIRecommendations` component to present these dynamic recommendations with detailed specifications, pricing, and visual cues.
-

Phase 11: UI/UX Enhancements & Homepage Modernization

1. Dynamic Banner Slider

- **Promo Banners:** Replaced legacy static promotional blocks on the homepage with an interactive, responsive slider displaying high-quality marketing banners.
 - **Sticky Sidebars:** Applied sticky layout positioning to filter sidebars on the search and catalog pages, improving navigation ergonomics on larger viewports.
-

Phase 12: Profile Decoupling & Security Hardening

1. Modular Profile Components

- **Refactoring:** Decoupled the monolithic `/profile` view into specialized, focused client components:
 - `profile-form.tsx`: Handles details updating.
 - `change-password-form.tsx`: Full flow to securely update user passwords.

- `profile-sidebar.tsx` & `profile-stats.tsx`: Standardized profile navigation and quick metrics dashboards.
 - `my-bids.tsx` & `order-history.tsx`: Dedicated lists for real-time tracking of active bids and past orders.
 - **Verification Scripts**: Added server-side checks (`check-functions.ts`, `check-rpc.ts`, `check-tables.ts`) to audit relational triggers and verify database health.
-

Phase 13: Seller Portal & Product Deletion

1. Authorized Product Deletion

- **Server Action**: Implemented a secure deletion action in `actions/product.ts` with checks to guarantee that only the product owner or system administrators can remove a listing.
- **Client Operations**: Integrated deletion options in admin and seller tables with visual confirmations.

2. Seller Auction Management

- **Dedicated Auction Views**: Developed a comprehensive detail page for sellers at `/seller/auctions/[id]` to monitor active bid streams, bidder profiles, and historical trends for their specific listings in real time.
-

Current Status

With Phase 3 complete, the platform boasts a highly modular dashboard, active AI assistance, advanced wishlist and seller tracking features, and robust authorized management pipelines.

Next Steps:

- Scale real-time notification workers for system-wide outbid events.
- Implement automated vendor payout flows.